MIL-P-20444C (PA)
28 February 1974
SUPERSEDING
MIL-P-20444B (MU)
12 August, 1963 and
MIL-P-46213A (MU)
6 September 1963
(see 6.10)

MILITARY SPECIFICATION

PRIMER, PERCUSSION, M42
PARTS FOR AND LOADING, ASSEMBLING AND PACKING

*This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 This specification contains quality assurance provisions and special requirements for the parts, loading, assembling and packing of one type of initiating device designated as Primer, Percussion, M42.

2. APPLICABLE DOCUMENTS

2.1 The following documents of the issue in effect on date of invitation for bids, or request for proposals, form a part of this specification to the extent specified herein.

SPECIFICATIONS

MILITARY

MIL-A-2550 - Ammunition, General Specification For

STANDARDS

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection by Attributes (ABC-STD-105)

FSC: 1375

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MIL-STD-109 - Quality Assurance Terms and Definitions
MIL-STD-1168- Lot Numbering of Ammunition
MIL-STD-1235- Single and Multilevel Continuous Sampling Procedures and Tables for Inspection by Attributes

*DRAWINGS

U.S. ARMY PICATINNY ARSENAL

PRODUCT AND PACKING DRAWINGS

7548347 - Box, Set-up, Packing, Ammo, Outer, for Relays or Primers, Assembly, Details and Marking.
7548388 - Box, Packing, Ammo, for Relay, M4, M5, M7 or Primer, M42, Assembly, Details and Packing
7548389 - Box, Set-up, Packing, Ammo, Inner, for Relays or Primers, Assembly Details, Packing and Marking.
8799925 - Primer, Percussion, M42, Assembly

INSPECTION EQUIPMENT DRAWINGS

8815991 - Holder, Primer
8815992 - Pin, Firing
8816003 - Adapter, Primer Holder
8816014 - Adapter, Ball

(Copies of specifications, standards, and drawings required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the Contracting Officer).

*3. REQUIREMENTS

3.1 Material. - Materials shall be in accordance with applicable drawings and specifications.

3.2 Primer. - The primer shall comply with all requirements specified on Drawing (dwg) 8799925 and with all requirements specified in applicable specifications.

3.3 Functioning (see 6.5). - The primer shall function under an applied energy of 25.49 inch ounces, maximum, without failure or

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evidence of squibbing or hangfire. (See 6.6) The primer shall not function under an applied energy of 3.84 inch ounces, minimum.

3.4 Workmanship. - The primers shall be loaded and assembled in a thorough, workmanlike manner. All parts shall be free of burrs, chips, sharp edges, dirt, grease, corrosion products and other foreign matter. The cleaning method used shall not be injurious to any part nor shall the parts be contaminated by the cleaning agent. All required marking shall be neat and sharply defined.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.-Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract or order, the supplier may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements. Reference shall be made to MIL-STD-109 to define terms used herein. The provisions of MIL-A-2550 shall apply.

4.1.1 Submission of Product.-At the time each completed lot of items deliverable under the contract is submitted to the Government for acceptance, the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product being submitted.

a. A statement that the lot complies with all of the quality assurance provisions specified in this specification.
b. Drawing and specification number and date, together with identification and date of changes thereto.
c. A statement that all material purchased by the contractor meets requirements, when such material is controlled by Government or commercial specifications referenced in any of the contractual documents, and that certificates of conformance are on file and available for review.
d. Number of items in the lot.
e. Date submitted.

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The certificate shall be signed by a responsible agent of the certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agents authority will not be required with subsequent certificates unless, during the course of the contract, this authority is vested in another agent of the certifying organization.

4.2 Inspection Provisions

4.2.1 Lot formation.-The term "inspection lot" as used in this specification is defined as an essentially homogeneous collection of units of product from which a representative sample is drawn and inspected to determine conformance with applicable requirements. The sample selected shall represent only that quantity of units from which the sample was drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer in one unchanged process using the same materials and methods in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions and complies with the provisions for submission of product as specified in MIL-STD-105. Changes to the process, specifications, or drawings not affecting safety, performance, interchangeability, or storage, as determined by the Government, shall not be deemed to alter the homogeneity of an inspection lot. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein regardless of the type of sampling procedure which is being applied to determine conformance with requirements. Lot numbering, as required, shall be in accordance with MIL-STD-1168.

*4.2.2 Examination.-The examinations listed in the following subparagraphs shall be performed on inspection lots (see 4.2.1) of the components or assemblies designated in the applicable paragraph headings. Inspection for critical defects (and major defects, when so specified) shall be 100 percent. Sampling plans for major and minor defects shall be in accordance with MIL-STD-105 except that continuous sampling plans in accordance with MIL-STD-1235 may be used if approved by the Government. Instead of applying the AQL's designated herein to the groups of major and minor defects listed, the contractor may elect to apply AQL's to individual defects using an AQL of 0.40 percent for each

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major defect and an AQL of 0.65 percent for each minor defect. When this method is elected, it shall be applied to all of the major and minor defects listed in the applicable sub-paragraph of this section, except where 100 percent inspection is specified, and be documented as part of the contractor's inspection system. Equipment necessary for the performance of the inspections listed shall be in accordance with 4.3.4.

4.2.2.1 Cup (See dwg. 8837991 covering a detail of dwg. 8799925).

Categories	Defects	Method of Inspection	Code No. (see 6.2)
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Thickness through bottom	Gage	01001
102.	Total length, minimum (min) ...	Gage	01002
103.	Cup damaged	Visual	01003
Minor:	AQL 1.50 percent		
201.	Inside diameter, min.	Gage	01004
202.	Outside diameter, maximum (max)	Gage	01005
203.	Evidence of poor workmanship (see 3.4)	Visual	01006

4.2.2.2 Anvil (see dwg. 8837992 covering a detail of dwg. 8799925).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Two or more vents partially or completely closed	Visual	02001
2.	Anvil malformed	Visual	02002
Major:	AQL 0.65 percent		
101.	Outside diameter	Gage	02003
102.	Total length	Gage	02004
103.	Blunt point	Visual	02005
104.	Lamination or chip	Visual	02006
Minor:	AQL 1.50 percent		
201.	Rings on face	Visual	02007

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202.	Unsymmetrical vent hole	Visual	02008
203.	Evidence of poor workmanship (see 3.4)	Visual	02009

4.2.2.3 Primer, Prior to Assembly of Anvil (see dwg. 8799925).

Categories	Defects	Method of Inspection	Code No.
Critical: None defined			
Major:	AQL 0.40 percent		
101.	Double or multiple cover	Visual	03001
102.	Cover missing	Visual	03002
Minor:	AQL 1.00 percent		
201.	Cover damaged or improperly inserted	Visual	03003
202.	Evidence of poor workmanship (see 3.4)	Visual	03004

4.2.2.4 Assembly (see dwg. 8799925).

Categories	Defects	Method of Inspection	Code No.
Critical:			
1.	Cup cracked or split	Visual	04001
Major:	AQL 1.00 percent		
101.	Flash hole obstructed	Visual	04002
102.	Belled cup	Visual	04003
103.	Anvil loose, cocked, damaged ..	Visual	04004
104.	Cup dented, flat on crown or otherwise malformed	Visual	04005
105.	Anvil below flush with cup	Visual	04006
106.	Heavy charge	Visual	04007
107.	Evidence of poor workmanship (see 3.4)	Visual	04008
Minor:	AQL 1.00 percent		
201.	Total length	Gage	04009
202.	Outside diameter	Gage	04010

4.2.2.5 Inner Set-up Box, prior to Closing (see dwg. 7548347).

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Categories	Defects	Method of Inspection	code No.
Critical:	None defined		
Major:	AQL 0.65 percent		
101.	Spacer damaged	Visual	05001
102.	Primer missing	Visual	05002
103.	Cushion missing or incomplete .	Visual	05003
Minor:	AQL 1.00 percent		
201.	Evidence of moisture	Visual	05004
202.	Evidence of poor workmanship (see 3.4)	Visual	05005

4.2.2.6 Inner Set-up Box, Wrapped (see dwg. 7548389).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Wrapping incomplete, torn, cut, or otherwise damaged	Visual	06001
102.	Contents loose	Manual	06002
Minor:	AQL 1.50 percent		
201.	Label missing	Visual	06003
202.	Marking incorrect, incomplete or illegible	Visual	06004
203.	Evidence of poor workmanship (see 3.4)	Visual	06005

4.2.2.7 Outer Set-up Box, prior to Sealing (see dwg. 7548347).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Evidence of moisture	Visual	07001

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Minor:	AQL 0.65 percent		
201.	Evidence of poor workmanship (see 3.4)	Visual	07002

4.2.2.8 Outer Set-up Box, Sealed (see dwg. 7548347).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Sealing strip incomplete, torn or improperly positioned .	Visual	08001
102.	Box punctured, cut or torn	Visual	08002
Minor:	AQL 1.50 percent		
201.	Label missing	Visual	08003
202.	Marking incorrect, incomplete or illegible	Visual	08004
203.	Evidence of poor workmanship (see 3.4)	Visual	08005

4.2.2.9 Sealed Bag, when Applicable (see dwg. 7548347).

Categories	Defects	Method of Inspection	Code No.
Critical:	None defined		
Major:	AQL 0.40 percent		
101.	Bag improperly sealed	Visual	09001
102.	Bag punctured or torn	Visual	09002
Minor:	AQL 1.50 percent		
201.	Bag damaged (not punctured completely through)	Visual	09003
202.	Marking incorrect, incomplete or illegible	Visual	09004
203.	Evidence of poor workmanship (see 3.4)	Visual	09005

4.2.2.10 Packing Box (see dwg. 7548388).

Categories	Defects	Method of Inspection	Code No.

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Critical: None defined

Major:	AQL 1.00 percent		
101.	Box damaged	Visual	10001
102.	Strapping clip damaged	Visual	10002
103.	FSC number missing or incorrect	Visual	10003
104.	Board loose	Manual	10004
105.	Strapping missing, broken, or loose	Visual/ Manual	10005
Minor:	AQL 1.50 percett		
201.	Marking incorrect, incomplete or illegible	Visual	10006
202.	Strapping improperly assembled.	Visual	10007
203.	Contents loose	Manual	10008
204.	Evidence of poor workmanship (see 3.4)	Visual	10009

4.2.3 TESTING

4.2.3.1 Composition of PA-101 primer mixture (when applicable) (see dwg. 8799925) - Major Defect - Code No. 11001. - The composition of the primer mixture shall be determined for each batch or blending of batches. Three samples of approximately 6.0 grams each shall be selected from each batch or blend of batches and shall be analyzed as specified in 4.3.1. should the averaged values obtained for the three samples fail to meet one or more of the composition requirements specified on the drawing, the batch or blend of batches shall be rejected.

4.2.3.2 Functioning (see 3.3)

4.2.3.2.1 Sensitivity - Major Defect - Code No. 04011. - A sample of 600 percussion primers shall be selected from each lot for this test. If the sum of the average critical height (H average) plus 5 times the standard deviation(s) multiplied by the weight of the ball exceeds the maximum requirement, or if the sum of the average critical height (H average) minus 2 times the standard deviation(s) multiplied by the weight of the ball is less than the minimum requirement, the lot shall be rejected (see 6.4). The test shall be performed as specified in 4.3.2.1, using test equipment specified in 4.2.4 and 4.3.2.1.

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4.2.3.2.2 Duds - Major Defect - Code No. 04012. - A sample of 300 percussion primers shall be selected from each lot for this test. Failure to function of one or more primers during this test shall be cause for rejection of the entire lot. The test shall be performed as specified in 4.3.2.2 using test equipment specified in 4.2.4 and 4.3.2.

*4.2.4 Inspection Equipment. - Inspection equipment required to perform the inspection prescribed in this specification is identified in the Examination (4.2.2) and Test Method (4.3) paragraphs herein. See 6.3 for details concerning responsibilities for inspection equipment design and approval.

*4.2.5 Visual inspection of primer (see dwg. 8799925) - Major Defect - Code No. 04013. - Each primer in the lot shall be inspected 100 percent for the following defects. Any primer containing one or more such defects shall be classed defective and removed from the lot.

a. Anvil missing or improperly positioned.
b. Anvil malformed
c. Charge missing (prior to assembly of cover)
d. Double anvil
e. Foil stained prior to pressing in primer.

4.3 Test Methods and Procedures.

4.3.1 Composition of Primer Mixture. - The composition of the primer mixture shall be determined by performing the following tests in the sequence specified:

4.3.1.1 Barium Nitrate. - A sample of approximately 1 gram of primer mixture shall be selected and weighed to the nearest milligram. The sample shall be dried to a constant weight at 60 plus or minus 5 degrees C., or in the desiccator containing calcium chloride. After drying, the sample shall be transferred to a tared medium-porosity, fritted glass crucible of 20 milliliter (ml.) capacity. Three ml. of distilled water shall be added at a temperature of 5 plus or minus 2 degrees C., agitated for 1 minute, and, if necessary, lumps shall be broken up with a rubber policeman attached to a glass rod. Suction shall be applied. This procedure shall be repeated five times, making a total of six, 3-ml. extractions. The sample in the filtering crucible shall be rinsed three times with alcohol, dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes,

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cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of barium nitrate as follows:

Barium nitrate, percent = (A-B) divided by W

where:

A = weight of crucible and sample
B = weight of the crucible and sample after extractions with water
W = weight of sample

*4.3.1.2 Basic lead styphnate. - The residue obtained in the determination of barium nitrate shall be extracted with six 10 ml. portions of saturated ammonium acetate solution at about 25 degrees C., (hot saturated ammonium acetate shall not be used as it dissolves antimony sulfide to some extent) agitating each portion for about 30 seconds and allowing each portion to remain in contact with the residue for approximately 2 minutes. The residue shall be washed with water until the filtrate coming through the crucible appears colorless. The filtrate and washings shall be quantitatively transferred to a 250-ml. calibrated volumetric flask, the beaker washed thoroughly with distilled water and diluted to the mark. Using a calibrated pipette, a 2-ml. aliquot of this solution shall be tansferred to a 50-ml. calibrated volumetric flask and diluted to the mark with distilled water. A portion of this solution shall be transferred to a corex glass spectrophotometric cell having a width of approximately 1 centimeter (cm.) and its optical density determined at a wave length of 410 millimicrons. As reference, a similar cell containing ammonium acetate solution of the same concentration as the sample solution shall be used. Since the cell which holds the basic lead styphnate is not identical with the reference cell, the optical density shall be corrected for the difference in the amount of light which the two cells scatter and absorb. To do this, both cells shall be filled with ammonium acetate solution and the optical density of the cell which originally contained the basic lead styphnate solution shall be measured at a wave length of 410 millimicrons. The percentage of basic lead styphnate in the sample shall be calculated as follows:

Basic lead styphnate, percent = 28 (C-D) divided by WE

where:

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C = absorbance of basic lead styphnate solution
D = absorbance of ammonium acetate solution
E = width of corex cell, cm.
W = weight of sample

4.3.1.3 Tetracene. - The wet residue obtained in the determination of lead styphnate shall be washed three times with alcohol, dried first in air until the odor of alcohol is negligible, and then in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The dried residue in the crucible shall be tansferred to a 125-ml. beaker with a stream of water, 25-ml. of water shall be added, and the contents of the beaker boiled on a hot plate for 5 minutes. The contents of the beaker shall be poured into the same crucible, and the residue in the crucible washed three times with boiling water, and finally with alcohol. The crucible and contents shall be dried in an oven maintained at 60 plus or minus 5 degrees C. for 30 minutes, cooled in a desiccator and weighed. The loss in weight shall be calculated as the percentage of tetracene in the sample:

Tetracene, percent = (F-G) 100 divided by W

where:

F = weight of crucible and residue after extraction with saturated ammonium acetate solution
G = weight of crucible and residue after treating with hot water
W = weight of the sample

4.3.1.4 Aluminum. - The dry residue obtained in the determination of tetracene shall be carefully transferred to a beaker. Carefully add 25 ml. of 50-percent hydrochloric acid solution and cover the beaker with a watch glass. After the reaction has subsided, place the beaker and contents on a hot plate and boil for 5 minutes. Remove the beaker from the hot plate and allow to cool. Filter the contents of the beaker through a filtering crucible of medium porosity and wash the beaker and crucible with ten - 5 ml. portions of water, catching the filtrate and washings in a clean beaker. Add 5 gms. of ammmonium chloride and 5 ml. of concentrated nitric acid and heat the solution to boiling on a hot plate. Remove the beaker from the hot plate and allow to coool. Add 2-3 drops of methyl red indicator and then add dropwise ammonium hydroxide until the alkaline color is obtained. Heat the solution to boiling and filter

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through No. 41 Whatman filter paper. Wash the precipitate several times with a 2-percent ammonium nitrate solution, then transfer the filter paper and precipitate to an accurately weighed porcelain or platinum crucible. The crucible and contents shall be dried and then ignited in a muffle furnace, gently at first, until the paper is thoroughly charred, and then gradually increase the heat. The crucible shall be covered and heated at 900-1000 degrees C. for one hour, cooled in a desiccator, and weighed immediately upon removal from the desiccator. The percent aluminum content shall be calcualted as follows:

Aluminum, percent = (52.91) (H-I) divided by W

where:
H = weight of crucible and preciptate
I = weight of crucible
W = weight of sample

4.3.1.5 Antimony sulfide. - The antimony sulfide content in the sample on a dry basis, shall be determined by subtracting from 100 the combined percentages of barium nitrate, basic lead styphnate, tetracene and aluminum.

*4.3.2 Functioning. - Prior to performing these tests the primers shall be conditioned at a temperature of 75 degrees Fahrenheit (F) plus or minus 5 degrees F., for a period of 16 hours, minimum. Method of inspection shall be in accordance with 8815991, 8815992, 8816003 and 8816014.

4.3.2.1 Sensitivity. - The sample of 600 primers shall be divided into 12 groups of 50 primers each and tested in the following manner:

The first group (50) of primers shall be tested by dropping a steel ball 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of 7 inches. (The height of drop shall be measured as the height from the top of the firing pin to the bottom of the ball). The number of primers that fire, misfire, hangfire or squib (see 6.6) shall be recorded. This procedure shall be repeated at successively increasing increments (see 6.7) of 1 inch, using a new group of 50 primers at each different height, until a height is reached at which all primers in the group being tested fire. The test fixture shall then be set at a height of 6 inches and the above procedure repeated

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at successively decreasing heights of 1 inch using a new group of 50 primers at each height, until a height is reached at which all primers in the group being tested fail to fire. Acceptance or rejection of the lot shall be determined from calculations based on recorded data (see 6.4).

NOTE: All samples subjected to this test shall be scrapped. Any remaining untested samples may be returned to the lot from which they were taken.

4.3.2.2 Duds. - Each primer of the sample of 300 primers shall be tested by dropping a steel ball weighing 1.94 plus or minus .02 ounces on the firing pin of the test fixture from a height of 13 inches.

5. PREPRATION FOR DELIVERY

5.1 Preservation and Packaging.

5.1.1 Levels B (see 6.1) and C. - The primers shall be preserved and packaged in accordance with dwgs. 7548347 and 7548389.

5.2 Packing.

5.2.1 Levels B (see 6.1) and C. - The packaged primers shall be packed in accordance with dwg. 7548388.

5.3 Marking. - Marking shall be as specified on the applicable packaging and packing drawings.

*6. NOTES

6.1 Ordering data.-Procurement documents shall specify the following:

a. Title, number and date of this specification.
b. Data cards. - Data cards shall be prepared for each lot in accordance with the information specified in Standard MIL-STD-1167. The cards shall contain the dud testing height and the sensitivity level obtained in 4.3.2.
c. Applicable level of packaging and packing. (The procuring activity shall specify level B, use of the barrier bag if the expected storage time will be in excess of 180 days).

6.2 Inspection code numbers.-The five digit code numbers assigned to the inspections herein are to facilitate future

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data collection and analysis by the Government. They should be cited as references on drawings of equipment designs submitted by the contractor to the Government for approval (see 6.3.3).

6.3 Inspection Equipment Designs.-Inspection equipment designs are of two types - Government Special Inspection Equipment (SIE) designs and contractor designs. SIE designs are designated by drawing numbers under the "Method of Inspection" headings in Section 4. Design responsibility for all other inspection equipment is assigned to the contractor. The contractor need not furnish any design when a complete Government SIE design is provided. Unless otherwise specified, however, the contractor may submit alternate or modified contractor designs of SIE in accordance with 6.3.2 and 6.3.3 should he elect to do so.

6.3.1 SIE designs.-SIE designs may consist of any of the following:

a. Detailed drawings which completely depict all information necessary for the fabrication and use of the item of inspection equipment.

b. A source control drawing or a specification control drawing as defined in MIL-STD-100.

c. An envelope drawing, as defined in MIL-STD-100, which establishes the criteria which a detailed design shall meet. When envelope drawings are specified, the contractor shall prepare designs which comply with the criteria therein.

6.3.2 Contractor designs.-Contractor designs are required for all inspection equipment for which SIE designs are not specified and may include commercial equipment which the contractor proposes to use. (Commercial equipment is defined as unmodified equipment which is cataloged and available for purchase by the general public). Contractor designs shall include appropriate operating instructions, calibration procedures and maintenance procedures. Commercial equipment shall be fully described by catalog listings or other means which provide sufficient information to permit identification and evaluation by the Government and may include illustrations and engineering data. Designs shall be prepared for any special fixture(s) required to be used with commercial equipment, or with SIE designs if not otherwise covered thereby (see 6.3.1c). Designs shall be of the category and form (per MIL-D-1000) specified in the Contract Data Requirements Lists

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(DD Form 1423). The contractor is referred to MIL-HDBK-204, "Inspection Equipment Design" for guidance. The specification number and the applicable five digit defect code number (or other specific identifying information) from Section 4 of this specification shall be referenced on each contractor design together with the component or assembly drawing number and revision letter to which the specific design applies.

6.3.3 Submission of Designs for Approval.-Contractor designs shall be approved by the Government prior to fabricating or procuring the equipment. Designs shall be submitted for approval to Picatinny Arsenal, ATTN: SARPA-QA-A-M, Dover, New Jersey 07801 in accordance with the stipulations, time frame and distribution specified in the Contract Data Requirements List (DD Form 1423) or in the contract. The contractor must include in his submission of designs a listing of any and all Government SIE designs which will be utilized. Partial submission of inspection designs is permissible and encouraged. However, the completion date for design review will be based on the date of the final submission of designs. Picatinny Arsenal design review will normally be accomplished within one month after receipt.

6.4 Formula and Method of determing H average and s.

6.4.1 Formula:

H average = A (plus) B (plus) C
s = I times the square root of L (minus) M.

where:

A = the sum of column 3.
B = the first drop height at which all elements misfired
C = one-half the drop height increment (see 6.7).
L = the sum of Column 5.
M = the square of the sum of Column 3.
I = the drop height increment.

6.4.2 Tabulation of data. - Divide a sheet of paper into 5 columns and tabulate the data as follows:

a. Title column 1 "Height of Drop" and list all drop heights in consecutive order starting with the lowest height. DO NOT list the heights at which all elements fired or misfired.

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b. Title column 2 "Number of Failures" and list the sum of the number of elements that misfired, squibbed or hang fired (see 6.6) at each intermediate height.

c. Title Column 3 "Decimal Fraction of Elements Failing to Fire at Each Intermediate Height" and record the decimal fraction that failed to fire at each intermediate height. This number is obtained by dividing the value obtained under "Number of Failures" by the number of elements tested at that height. Calculations should be carried out and recorded to the nearest one-hundredth (.01).

d. Title Column 4 "Standard Deviation Factor" and list odd numerals in sequence starting with 1, for as many drop heights as are listed in Column 1.

e. Title Column 5 "Standard Deviation Factor Times Fraction of Failures" and, by multiplying each decimal fraction column 3 by the number on the same line in Column 4, enter the product on the corresponding line in Column 5.

6.4.3 Procedure for solving equations:

a. To obtain "A" - Add the values in Column 3 and substitute the sum for "A".

b. To obtain "B". - The first drop height at which all primers misfired.

c. To obtain "C" - Divide the drop height increment by 2 and substitute the dividend for "C".

d. To obtain "L" - Add the values in Column 5 and substitute the sum for "L".

e. To obtain "M" - Square the sum of Column 3 and substitute the sum for "M"

f. To obtain "H average" - Add the values substituted for A, B and C.

g. To obtain "s" - Substract the value substituted for "M" from the value substituted for "L" and then obtain the square root of the difference, and then multiply the square root by the height of increment.

h. Using the calculated values for "H average" and "s", if "H average" plus 5 "s" multiplied by the maximum weight of the ball is equal to or less than the maximum inch ounce energy value required, and "H average" minus 2 "s" multiplied by the minimum weight of the ball is equal to or greater than the minimum energy value required, the lot should be considered acceptable for sensitivity.

6.5 Inch Ounces. - Inch ounces are defined as the drop height in inches multiplied by the weight of the ball in ounces.

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6.6 Failure. - Defined as any one of the following primer malfunctions:

a. Misfire - a failure wherein initiation of the primer mixture does not occur.

b. Squib - a failure wherein a burning of the primer mixture occurs without detonation.

c. Hangfire - A failure wherein an audible delay occurs between the instant of application of energy and the detonation report.

6.7 Height of Increment. - The height of increment is defined as a constant interval in inches between successive drop heights which must be considered in calculating average height and the standard deviation in drop height.

6.8 Check test for possible deterioration. - The following test is designed to prevent the loading of deteriorated primers into major ammunition items. The test should be performed on primers from each lot which have been stored under normal conditions for more than two years or under adverse conditions for any period of time. If the specification for the ammunition item into which the primers are loaded does not require that the check test be performed, the contracting officer should insert provisions for the performance of the check tests in the contract. The contracting officer should arrange to have the contractor reimbursed for any expenses incurred in the performance of the check test for possible deterioration.

6.8.1 Sensitivity. - The test shall be performed as specified in 4.3.2.1, using test equipment specified in 4.2.4. A sample of 900 primers shall be selected from each lot for this test. Seventy-five primers shall be tested at each height. If H average plus 4.66 s exceeds 13 inches, or if H average minus 1.86 s is less than 2 inches the lot shall be rejected.

6.8.2 Duds. - The test shall be performed as specified in 4.3.2.2, using test equipment specified in 4.2.4. A sample of 300 primers shall be selected from each lot for the test. Failure to function of 1 or more primers shall reject the lot.

*6.9 The margins of this specification are marked with an asterisk to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no

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liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

*6.10 Supersession Data. - This specification includes the requirements of MIL-P-46213A (MU) dated 6 September 1963.

Custodian:
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